

Project Design Phase-II Data Flow Diagram & User Stories

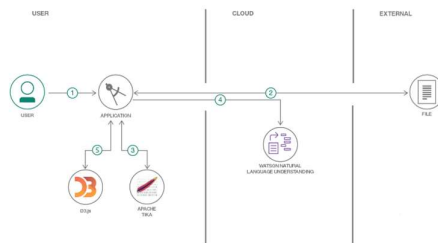
Date	10 July 2026
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

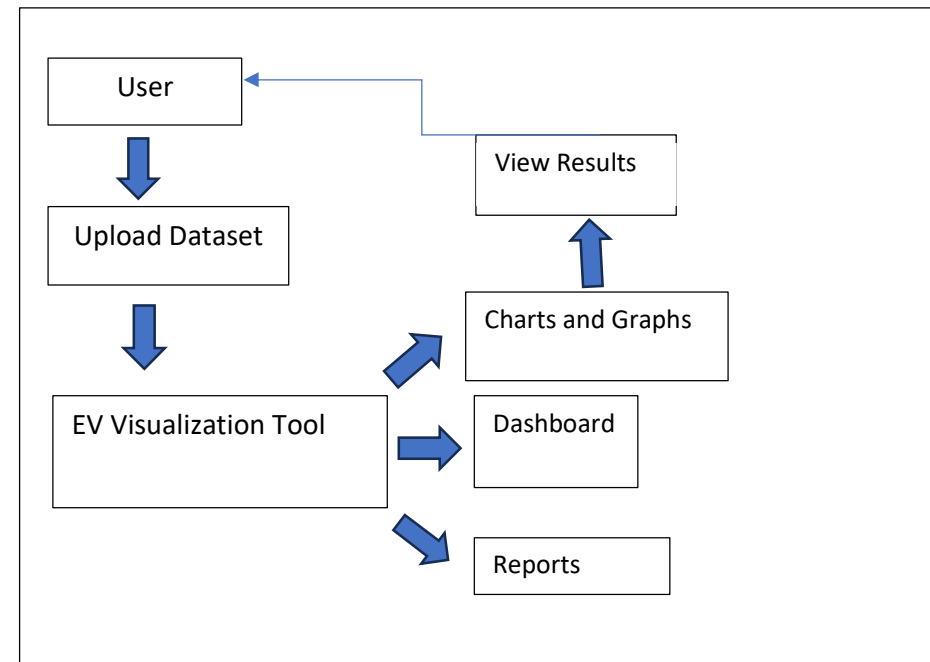
Example: (Simplified)

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

Example: DFD Level 0 (Industry Standard)



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
User	Dataset Upload	USN-1	As a user, I can upload an EV dataset for analysis.	Dataset uploads successfully.	High	Sprint-1
User	Dashboard	USN-2	As a user, I can view an interactive dashboard.	Dashboard displays correctly.	High	Sprint-1
User	Filtering	USN-3	As a user, I can filter data by vehicle brand.	Selected brand data is displayed.	High	Sprint-1
User	Filtering	USN-4	As a user, I can filter vehicles by model year.	Correct vehicles are shown.	High	Sprint-1
User	Visualization	USN-5	As a user, I can view bar charts comparing EV brands.	Bar chart is generated correctly.	High	Sprint-1
User	Visualization	USN-6	As a user, I can view pie charts showing vehicle types.	Pie chart displays accurate percentages	Medium	Sprint-2
User	Visualization	USN-7	As a user, I can compare electric range across vehicles.	Comparison chart is displayed.	High	Sprint-2
User	Search	USN-8	As a user, I can search for a specific EV model.	Search results are accurate.	Medium	Sprint-2
User	Report	USN-9	As a user, I can download dashboard reports.	Report downloads successfully.	Medium	Sprint-3
Administrator	Data Management	USN-10	As an administrator, I can update the EV dataset.	Updated data is reflected in the dashboard.	High	Sprint-3